

>> Exponential and Logarithmic Functions	Unit	CHECKPOINT		
		1	2	3
2A.2 Attributes of functions and their inverses. The student applies mathematical processes to understand that functions have distinct key attributes and understand the relationship between a function and its inverse. 2A.5 Exponential and logarithmic functions and equations. The student applies mathematical processes to understand that exponential and logarithmic functions can be used to model situations and solve problems. 2A.8 Data. The student applies mathematical processes to analyze data, select appropriate models, write corresponding functions, and make predictions. Connected Knowledge and Skills 2A.7				

Process (Tools to Know)	Unit	CHECKPOINT		
		1	2	3
2A.1(A) apply math in everyday situations ☉ 2A.1(B) use problem-solving models ☉ connected 2A.1(C)				

Content	Unit	CHECKPOINT		
		1	2	3
Inverse Functions				
2A.2(C) describe and analyze the relationship between a function and its inverse (quadratic and square root, logarithmic and exponential), including the restriction(s) on domain, which will restrict its range		Data included in "Quadratic and Square Root Functions"		
2A.2(B) graph and write the inverse of a function using notation such as $f^{-1}(x)$				
2A.5(C) rewrite exponential equations as their corresponding logarithmic equations and logarithmic equations as their corresponding exponential equations				

Exponential Functions				
2A.2(A) graph the functions $f(x)=\sqrt{x}$, $f(x)=1/x$, $f(x)=x^3$, $f(x)=\sqrt[3]{x}$, $f(x)=b^x$, $f(x)= x $, and $f(x)=\log_b(x)$ where b is 2, 10, and e , and, when applicable, analyze the key attributes such as domain, range, intercepts, symmetries, asymptotic behavior, and maximum and minimum given an interval		Data included in "Absolute Value Functions"		
2A.5(A) determine the effects on the key attributes on the graphs of $f(x) = b^x$ and $f(x) = \log_b(x)$ where b is 2, 10, and e when $f(x)$ is replaced by $af(x)$, $f(x) + d$, and $f(x - c)$ for specific positive and negative real values of a , c , and d				
2A.5(D) solve exponential equations of the form $y = ab^x$ where a is a nonzero real number and b is greater than zero and not equal to one and single logarithmic equations having real solutions				
2A.8(C) predict and make decisions and critical judgments from a given set of data using linear, quadratic, and exponential models ☉		Data included in "Quadratic and Square Root Functions"		
2A.5(B) formulate exponential and logarithmic equations that model real-world situations, including exponential relationships written in recursive notation				
2A.7(I) write the domain and range of a function in interval notation, inequalities, and set notation		Data included in "Absolute Value Functions"		
2A.8(A) analyze data to select the appropriate model from among linear, quadratic, and exponential models				
2A.8(B) use regression methods available through technology to write a linear function, a quadratic function, and an exponential function from a given set of data		Data included in "Quadratic and Square Root Functions"		

>> TEKS clusters typically requiring additional time and focus in the curriculum

(continued)

>> Exponential and Logarithmic Functions (continued)	Unit	CHECKPOINT		
		1	2	3
2A.2 Attributes of functions and their inverses. The student applies mathematical processes to understand that functions have distinct key attributes and understand the relationship between a function and its inverse. 2A.5 Exponential and logarithmic functions and equations. The student applies mathematical processes to understand that exponential and logarithmic functions can be used to model situations and solve problems. 2A.8 Data. The student applies mathematical processes to analyze data, select appropriate models, write corresponding functions, and make predictions. Connected Knowledge and Skills 2A.7				

Content	Unit	CHECKPOINT		
		1	2	3
Logarithmic Functions				
2A.2(A) graph the functions $f(x)=\sqrt{x}$, $f(x)=1/x$, $f(x)=x^3$, $f(x)=\sqrt[3]{x}$, $f(x)=b^x$, $f(x)= x $, and $f(x)=\log_b(x)$ where b is 2, 10, and e , and, when applicable, analyze the key attributes such as domain, range, intercepts, symmetries, asymptotic behavior, and maximum and minimum given an interval		Data included in "Absolute Value Functions"		
2A.5(A) determine the effects on the key attributes on the graphs of $f(x) = b^x$ and $f(x) = \log_b(x)$ where b is 2, 10, and e when $f(x)$ is replaced by $af(x)$, $f(x) + d$, and $f(x - c)$ for specific positive and negative real values of a , c , and d				
2A.5(D) solve exponential equations of the form $y = ab^x$ where a is a nonzero real number and b is greater than zero and not equal to one and single logarithmic equations having real solutions				
2A.5(B) formulate exponential and logarithmic equations that model real-world situations, including exponential relationships written in recursive notation				
2A.5(E) determine the reasonableness of a solution to a logarithmic equation				
2A.7(I) write the domain and range of a function in interval notation, inequalities, and set notation		Data included in "Absolute Value Functions"		

Process (Ways to Show)	Unit	CHECKPOINT		
		1	2	3
2A.1(E) create representations 2A.1(F) analyze information ⓘ connected 2A.1(D), 2A.1(G)				

>> TEKS clusters typically requiring additional time and focus in the curriculum